SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence   COURSE CODE: CSA17

COURSE OUTCOME: CO2 – Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 50%   Duration: 30 Minutes   Questions: 20   Total Marks: 20

MCQ TEST – CORRECT ANSWERS WITH EXPLANATIONS

Code of Conduct: I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ____________________

1. In informed search strategies, some algorithms use only the heuristic value to decide which node to expand next. This approach focuses on selecting the node that appears closest to the goal without considering the cost already incurred. Which algorithm follows this principle?

A) Breadth First Search
B) Uniform Cost Search
C) Greedy Best-First Search
D) Depth First Search

Correct Answer: C) Greedy Best-First Search

Explanation: Greedy Best-First Search selects the node with the lowest heuristic value h(n), which estimates the remaining cost to the goal. It does not consider the cost already travelled.

2. A* search is widely used in pathfinding and graph traversal because it combines the actual cost from the start node and the estimated cost to the goal. What is the evaluation function used in A* search?

A) f(n) = g(n)
B) f(n) = h(n)
C) f(n) = g(n) + h(n)
D) f(n) = g(n) × h(n)

Correct Answer: C) f(n) = g(n) + h(n)

Explanation: A* combines g(n), the actual cost from the start, with h(n), the estimated cost to the goal. Therefore, f(n) = g(n) + h(n).

3. Heuristic functions play a crucial role in informed search by guiding the algorithm toward the goal. A good heuristic should not overestimate the actual cost and should improve search efficiency. What property ensures optimality in A* search?

A) Completeness
B) Admissibility
C) Depth limit
D) Backtracking

Correct Answer: B) Admissibility

Explanation: An admissible heuristic never overestimates the true cost to reach the goal. With the required conditions, this property allows A* to find an optimal solution.

4. Local search algorithms operate by starting with an initial solution and iteratively improving it based on a given objective function. Which of the following is a local search algorithm?

A) A* Search
B) Hill Climbing
C) Breadth First Search
D) Uniform Cost Search

Correct Answer: B) Hill Climbing

Explanation: Hill Climbing is a local search method. It repeatedly moves from the current state to a better neighboring state according to an evaluation function.

5. In adversarial search, agents compete against each other in a game environment. Which algorithm is commonly used for such decision-making?

A) Greedy Search
B) Minimax Algorithm
C) Backtracking
D) Uniform Search

Correct Answer: B) Minimax Algorithm

Explanation: Minimax is designed for competitive two-player games. It chooses a move that maximizes the player's outcome while assuming the opponent tries to minimize it.

6. A delivery robot selects paths based only on the estimated distance to the destination without considering obstacles already encountered. Which algorithm is the robot using?

A) A* Search
B) Greedy Best-First Search
C) Uniform Cost Search
D) Depth Limited Search

Correct Answer: B) Greedy Best-First Search

Explanation: Greedy Best-First Search uses only h(n), the estimated distance or cost to the goal. Because it ignores the path cost g(n), it may produce a suboptimal route.

7. A GPS navigation system calculates routes by combining the distance already travelled and an estimate of the remaining distance to the destination. Which algorithm best represents this approach?

A) Depth First Search
B) A* Search
C) Hill Climbing
D) Backtracking

Correct Answer: B) A* Search

Explanation: A* uses both the cost already travelled, g(n), and the estimated remaining cost, h(n). Its evaluation function is $f(n)=g(n)+h(n)$.

8. A university is scheduling exams for multiple subjects with constraints such as no overlapping exams and limited rooms. The solution must satisfy all constraints while assigning time slots. This problem is best modeled as:

A) Game Playing Problem
B) Constraint Satisfaction Problem
C) Heuristic Search Problem
D) Optimization Problem

Correct Answer: B) Constraint Satisfaction Problem

Explanation: A CSP consists of variables, domains, and constraints. Exam scheduling assigns time slots and rooms while satisfying restrictions such as no conflicts.

9. An AI-based chess program evaluates possible moves and anticipates the opponent's responses before making a decision. Which algorithm is being applied here?

A) Greedy Search
B) Hill Climbing
C) Minimax Algorithm
D) Breadth First Search

Correct Answer: C) Minimax Algorithm

Explanation: Chess is an adversarial game. Minimax examines the player's possible moves and the opponent's responses to select the best move.

10. A robot explores an unknown environment where it does not have prior knowledge of obstacles. It makes decisions step-by-step and updates its knowledge as it moves. Which type of agent is suitable?

A) Offline Agent
B) Online Search Agent
C) Static Agent
D) Deterministic Agent

Correct Answer: B) Online Search Agent

Explanation: An online search agent acts while discovering the environment. It does not require a complete map beforehand and updates its knowledge during exploration.

11. A* search guarantees optimal solutions only when certain conditions are satisfied by the heuristic function. Why is an admissible heuristic important?

A) It reduces memory usage
B) It ensures optimality
C) It increases branching factor
D) It avoids loops

Correct Answer: B) It ensures optimality

Explanation: An admissible heuristic never overestimates the actual remaining cost. This preserves the conditions required for A* to return an optimal solution.

12. In large game trees, evaluating every node becomes computationally expensive. Alpha-Beta pruning helps eliminate branches that do not affect the final decision. What is the primary benefit?

A) Reduces depth of tree
B) Reduces number of nodes evaluated
C) Changes game rules
D) Improves heuristic accuracy

Correct Answer: B) Reduces number of nodes evaluated

Explanation: Alpha-Beta pruning removes branches that cannot influence the final minimax decision. Therefore, fewer nodes need to be evaluated.

13. Hill Climbing continuously moves toward better states based on local improvements. What is the main drawback?

A) High memory usage
B) Revisiting nodes
C) Getting stuck in local maxima
D) Requires complete tree

Correct Answer: C) Getting stuck in local maxima

Explanation: Hill Climbing can reach a state that is better than its neighbors but worse than the global optimum. This is called a local maximum.

14. Constraint Satisfaction Problems involve variables, domains, and constraints. What ensures a valid CSP solution?

A) Heuristic function
B) Path cost
C) Constraint satisfaction
D) Depth-first traversal

Correct Answer: C) Constraint satisfaction

Explanation: A CSP solution is valid only when values are assigned to the variables and all specified constraints are satisfied.

15. In adversarial games, evaluation functions are used when it is not feasible to explore the entire game tree. What is the role of an evaluation function?

A) Generate successors
B) Estimate utility of a state
C) Reduce branching factor
D) Store moves

Correct Answer: B) Estimate utility of a state

Explanation: An evaluation function estimates how favorable a game state is when the complete game tree cannot be searched to terminal states.

16. In A* search, suppose a node has a path cost of 7 and a heuristic estimate of 5. What is the value of f(n)?

A) 12
B) 2
C) 35
D) 7

Correct Answer: A) 12

Explanation: Using f(n)=g(n)+h(n), we get f(n)=7+5=12.

17. A heuristic function sometimes overestimates the actual cost to reach the goal. This violates admissibility. What happens in such a case?

A) A* remains optimal
B) A* becomes non-optimal
C) A* becomes BFS
D) A* stops working

Correct Answer: B) A* becomes non-optimal

Explanation: If the heuristic overestimates the true cost, A* can discard or delay an optimal path. Therefore, optimality is no longer guaranteed.

18. Consider a search tree with a branching factor of 2 and depth of 4. The total number of nodes generated in a full binary tree can be calculated analytically. What is the approximate number of nodes?

A) 10
B) 15
C) 31
D) 16

Correct Answer: C) 31

Explanation: For a full binary tree through depth 4, total nodes = 1+2+4+8+16 = 31. Therefore, the answer is 31.

19. In a minimax tree, each node has 3 children and the depth is 3. How many leaf nodes are present?

A) 9
B) 27
C) 6
D) 12

Correct Answer: B) 27

Explanation: The number of leaves is b^d. Here b=3 and d=3, so 3^3=27 leaf nodes.

20. Alpha-Beta pruning improves the efficiency of minimax by reducing the number of nodes evaluated. What is the best-case time complexity?

A) O(b^d)
B) O(b^(d/2))
C) O(d^b)
D) O(log b)

Correct Answer: B) O(b^(d/2))

Explanation: In the best case, Alpha-Beta pruning can reduce the effective search depth approximately by half, giving O(b^(d/2)) instead of O(b^d).